

Q4 Exercise

1.(10%) To understand the cost/complexity/performance trade-offs of forwarding in a pipelined processor. Refer to pipelined datapaths from Figure 4.45, assume that, all the instructions executed in a processor, the following fraction of these instructions have a particular type of RAW data dependence. The type of RAW data dependence is identified by the stage that produces the result (EX or MEM) and the instruction that consumes the result (1st instruction that follows the one that produces the result, 2nd instruction that follows, or both). Assume that the register write is done in the first half of the clock cycle and that register reads are done in the second half of the cycle, so "EX to 3rd" and "MEM to 3rd" dependencies are not counted because they cannot result in data hazards. Also, assume that the CPI of the processor is 1 if there are no data hazards.

- "EX to 1st only 10%" indicates that the data produced in the EX stage is required only by the next instruction, and 10% represents that this scenario occurs in 10% of all executed instructions
- "EX to 2nd only 20%" indicates that the data produced in the EX stage is required only by the second instruction, and 20% represents that this scenario occurs in 20% of all executed instructions

EX to 1 st Only	MEM to 1 st Only	EX to 2 nd Only	MEM to 2 nd Only	EX to 1 st and MEM to 2 nd	Other RAW Dependences
10%	20%	5%	15%	10%	5%

1.1. (5%) If we use no forwarding, what fraction of cycles are we stalling due to data hazards?

No forwarding $\rightarrow$ 等 WB 才能讀

Type	Stall
EX 1 st	2
MEM 1 st	2
EX 2 nd	1
MEM 2 nd	1
Both	$\max(1, 2)$
Other	0

$$\begin{aligned}
 &10\% \times 2 \\
 &20\% \times 2 \\
 &5\% \times 1 \\
 &15\% \times 1 \\
 &+ 10\% \times 2 \\
 &\hline
 &100\%
 \end{aligned}$$

$$\begin{aligned}
 \text{Fraction} &= \frac{\text{Stall}}{\text{CPI} + \text{Stall}} \\
 &= \frac{1}{1+1} \\
 &= \frac{1}{2}
 \end{aligned}$$

A: 50%

1.2. (5%) If we use full forwarding (forward all results that can be forwarded), what fraction of cycles are we stalling due to data hazards?

Type	Stall
EX 1 st	0
MEM 1 st	1
EX 2 nd	0
MEM 2 nd	0
Both	0
Other	0

$$20\% \times 1 = 20\%$$

$$\frac{0.2}{1+0.2} = 0.1667$$

A: 16.67%

2. (10%) Given the operation times for the major functional units are: 100ps for memory access, 150ps for ALU operation; and 56ps for register file read or write. Assuming that all the other delays (like control unit, multiplexer, pipeline overheads, etc) are negligible. Assume only R-type, lw, sw are supported. Fill in the blank in (a) ~ (b)

(a) (5%) For a single-cycle CPU where the instruction with the longest latency determines the clock cycle time, the clock cycle time is 462 ps.

(b) (5%) It takes 4625 ps to execute the code sequence below using the single-cycle CPU

code sequence	
1	lw \$t1, 0(\$s1)
2	sw \$s1, 0(\$s2)
3	add \$t2, \$s2, \$s3
4	add \$t3, \$s1, \$s2
5	lw \$t1, 0(\$t2)

$$\begin{aligned}
 \text{lw} &: \text{IF} + \text{ID} + \text{EX} + \text{MEM} + \text{WB} \\
 &= 100 + 56 + 150 + 100 + 56 \\
 &= 462 \text{ ps} \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{sw} &: \text{IF} + \text{ID} + \text{EX} + \text{MEM} \\
 &= 406 \text{ ps}
 \end{aligned}$$

$$\begin{aligned}
 \text{add} &: \text{IF} + \text{ID} + \text{EX} + \text{WB} \\
 &= 362 \text{ ps}
 \end{aligned}$$

3. (15%) Given a processor implemented by a single-cycle control and datapath shown in Figure 1. Assume that the functional blocks adopted to implement the datapath have the following latencies in Table 1, and we need to design a processor that can support the MIPS instructions listed in Table 2.

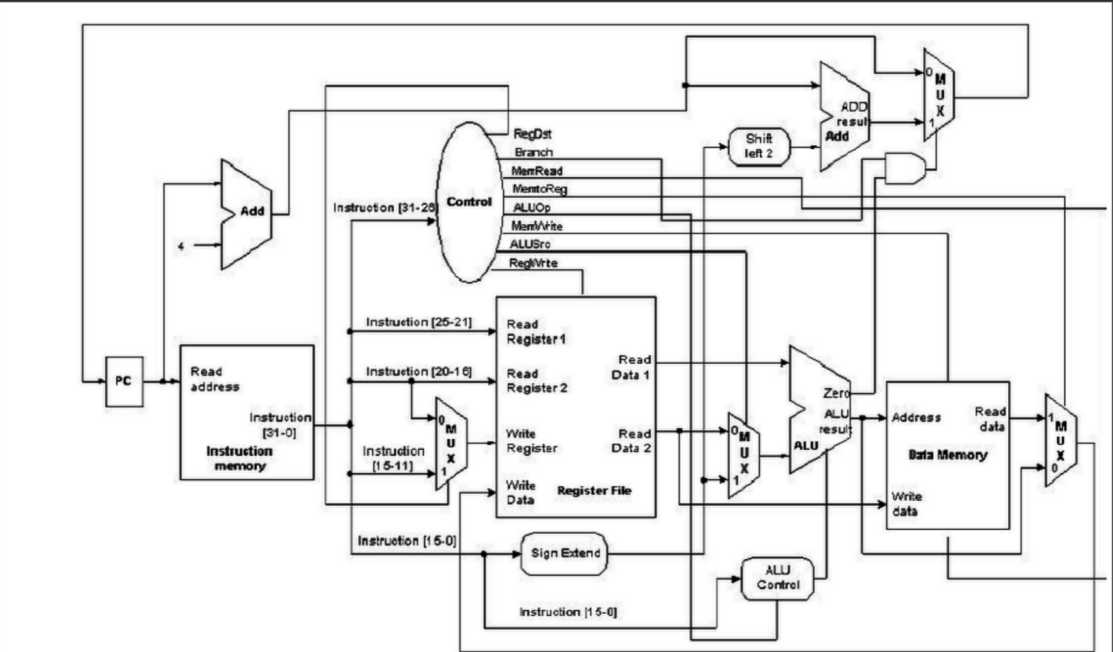


Figure 1. The single-cycle datapath with the control unit

PC's Clk.-to-Q	I-Mem.	Add	ALU	Ctrl. Unit	ALU-Ctrl.	Regs. Access	Data-Mem.	Sign-extend	AND-Gate	Shift-left-2	MUX delay
0ps	150ps	90ps	120ps	250ps	50ps	100ps	200ps	10ps	70ps	10ps	10 ps

Table 1. Latencies of the functional units on the datapath

Instr.	Opcode	Funct
add	0 ₁₀	32 ₁₀
sub	0 ₁₀	34 ₁₀
beq	4 ₁₀	
lw	35 ₁₀	
sw	43 ₁₀	

Table 2. MIPS Instruction Opcode & Funct fields

3.1(5%) If the only thing we need to do in a processor is fetch consecutive instructions (Figure 4.6), what would the cycle time be?

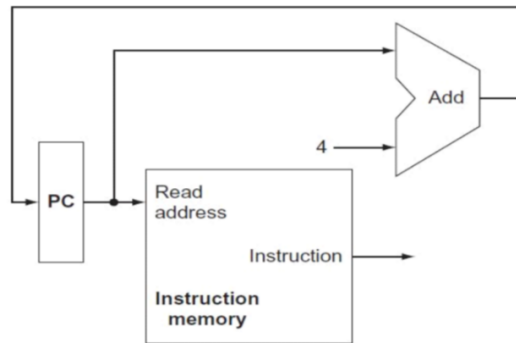


FIGURE 4.6 A portion of the datapath used for fetching instructions and incrementing the program counter. The fetched instruction is used by other parts of the datapath.

Ans: 150 ps

3.2(5%) If this processor only needs to support sw and beq instructions, please analyze the critical path of the required instructions, to draw the critical path and compute the clock cycle time.

beq : 650 ps.

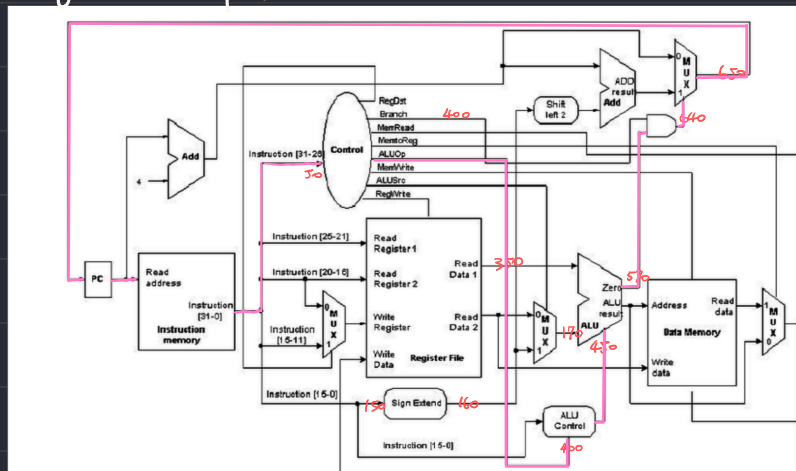


Figure 1. The single-cycle datapath with the control unit

sw : 770 ps

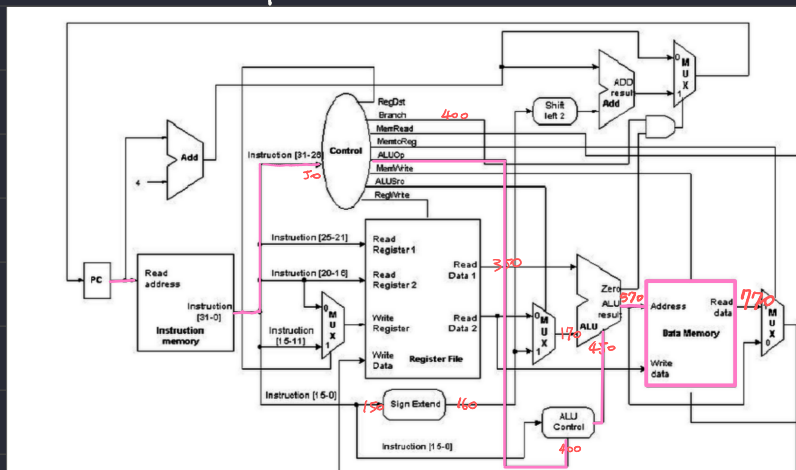


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